

Class 8 Sets

Goals for the class:

1. Become familiar with notation and terminology for sets.
2. Understand the connections between set theory and logic.
3. Know strategies to prove two sets are equal.

Notation

Typically denote sets by upper case A, B, X, Y
Elements in a set are denoted by lower case a is an element of set A .

I will use the letter U to indicate a prespecified universal set.

The symbol $\emptyset$ will indicate the set with no elements called empty set or null set

Intersection

$A \cap B$ = set of elements in both A and B .

Union

$A \cup B$ = set of elements in A or B or both.

Complement

A^c = set of elements in U but not in A .

Algebra of sets

The algebra of sets is exactly parallel to the algebra of sentential logic. It is another example of a Boolean algebra.

$A \cap B$ intersection $\longleftrightarrow p \wedge q$ conjunction (and)

$A \cup B$ union $\longleftrightarrow p \vee q$ disjunction (or)

A^c complement $\longleftrightarrow \sim p$ negation

DeMorgan

$$(A \cap B)^c = A^c \cup B^c \quad \sim(p \wedge q) \equiv (\sim p) \vee (\sim q)$$

Universal set U corresponds to tautology T

Empty set $\emptyset$ corresponds to contradiction F

Class Activity Write a set theory expression that corresponds to a line in the table below.

Formal Name	Logical Equivalence
Double negation	$\sim(\sim p) \equiv p$
De Morgan's laws	$\sim(p \wedge q) \equiv (\sim p) \vee (\sim q)$ $\sim(p \vee q) \equiv (\sim p) \wedge (\sim q)$
Implication expansion	$p \rightarrow q \equiv (\sim p) \vee q$
Contrapositive	$p \rightarrow q \equiv (\sim q) \rightarrow (\sim p)$
Biconditional expansion	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
Commutativity	$p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$
Associativity	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ $(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributivity	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Tautology	$p \vee (\sim p) \equiv T$
Contradiction	$p \wedge (\sim p) \equiv F$
Simplification	$p \wedge T \equiv p$ and $p \wedge F \equiv F$ $p \vee T \equiv T$ and $p \vee F \equiv p$

Proof Strategy: How to prove two sets are equal.

$A = B$ means every element of A is in B and every element of B is in A .

Notation $A \subseteq B$ and $B \subseteq A$

Example: Prove $(A \cup B)^c = A^c \cap B^c$

Proof: First show every element of $(A \cup B)^c$ is also in $A^c \cap B^c$.

Let $x \in (A \cup B)^c$.

This means x is not in $A \cup B$.

This means x is in neither A nor in B .

This means x is not in A and x is not in B .

This means x is in A^c and x is in B^c .

This means x is in both A^c and B^c .

So x is in $A^c \cap B^c$.

Therefore $(A \cup B)^c$ is contained in $A^c \cap B^c$.

Now go the other direction to show every element of $A^c \cap B^c$ is also in $(A \cup B)^c$.

Let $x \in A^c \cap B^c$ and reverse the previous steps.

Disprove the statement: If $A \not\subseteq B$ and $B \not\subseteq C$, then $A \not\subseteq C$.

Give counterexample

Let $A = \{1\}$, $B = \{2\}$, $C = \{1, 3\}$

$A \not\subseteq B$ and $B \not\subseteq C$ But $A \subseteq C$.

Other ways to create new sets from given sets besides intersection, union, and complement.

Difference of two sets

$$A \setminus B = A \cap B^c = \{x : x \in A \text{ and } x \notin B\}$$

Cartesian Product

$$A \times B = \text{set of ordered pairs} = \{(a, b) : a \in A \text{ and } b \in B\}$$

Power set

$$\mathcal{P}(A) = \text{set of all subsets of } A$$

Examples: Let $A = \{1, 3, 5, 7, 9\}$, $B = \{2, 3, 5, 7\}$ and $C = \{1, 2, 3\}$.

$$A \setminus B = \{1, 9\}$$

$$B \setminus A = \{2\}$$

$$B \times C = \{(2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), \\ (5, 1), (5, 2), (5, 3), (7, 1), (7, 2), (7, 3)\}$$

$$\mathcal{P}(C) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$